Geographic Information Systems

Custom CRS for historic "site" grid

Asked 2 years, 8 months ago Modified 1 year, 10 months ago Viewed 585 times

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I have a set of historic drawings which show a local site grid. I'd like to setup a custom CRS for this grid in QGIS so i can convert layers to/from this grid and MGA94 Zone 56 (EPSG:28356).

Presumably the drawing is shown with north to the top of the page, but after georeferencing it's rotated approx 9.4 deg clockwise.

I've read some grid intersections points from the drawing

Point 1: local xy = 7950, 1000 MGA94 Zone 56 = 383276.37, 6359888.50

Point 2: local xy = 7650, 850 MGA94 Zone 56 = 382941.47, 6359791.40

Point 3: Local xy = 7850, 850 MGA94 Zone 56 = 383155.14, 6359756.31

I've spent the day trying to wrap my head around:

- Reprojecting local CRS using PROJ
- WKT for local mine grid and
- Define custom CRS in WKT from point and angle

but can't make anything work. Can anyone help me out?

qgis coordinate-system proj

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edited Jul 2, 2020 at 8:43

user2856

58.7k 6 103 181

asked Jul 2, 2020 at 5:26

timtim

53 4

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First, some previous considerations and decisions: Both systems are 2D. EPSG:28356 is a projected (Transverse Mercator method) one. About the local reference system we don't know its procedence, but we can think that it is just a local cartesian 2D system.

We have some options, but let me consider two: One is transform the EPSG:28356 coordinates to geocentric, determine the paremeters of a similitude transformation from the local cartesian system to the geocentric one, in 3D, and then transform the coordinates of the local cartesian system to geocentric and reproject them to EPSG:28356. The other is consider a 2D similitude transformation between the local cartesian system and the projected one. This method will be less precise but has an advantage, we can define a CRS WKT for the 2D affine transformed from the projected reference system, and we can just define that CRS for the local layer.

Since you want to setup a custom CRS we will take the second option. So, lets find the 2D parameters to transform the coordinates from EPSG:28356 to the local CRS.

I will use a Python module that I wrote when I need to do that (https://github.com/gabriel-de-luca/simil). It is made for 3D, so I will create a Z=zero coordinate. There are other ways to get the parameters, but I use this:

```
import numpy as np
np.set_printoptions(precision=3,suppress=True)
import simil

source_points = [[383276.37, 6359888.50, 0],
                 [382941.47, 6359791.40, 0],
                 [383155.14, 6359756.31, 0]]

target_points = [[7950, 1000, 0],
                 [7650, 850, 0],
                 [7850, 850, 0]]

source_points_array = np.array(source_points)
target_points_array = np.array(target_points)

# Get the parameters
m_scalar, r_matrix, t_vector = simil.process(source_points_array,target_points_array)

# Print the parameters
print('\n m scalar = \n' + str(m_scalar))
print('\n R matrix = \n' + str(r_matrix))
```

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```
 m scalar =
0.9598327695807208

 R matrix =
[[ 0.985 -0.172  0.   ]
 [ 0.172  0.985  0.   ]
 [ 0.     0.     1.   ]]

 T vector =
[[  694357.794]
 [-6075863.374]
 [       0.   ]]

 A0 = [694357.794]

 A1 = 0.9455596465082436

 A2 = -0.1649117959886513

 B0 = [-6075863.374]

 B1 = 0.1649117959886513

 B2 = 0.9455596465082436
```

Now, we can create a custom CRS WKT2:2019 definition, which will be a derived (Affine method) from EPSG:28356:

```
DERIVEDPROJCRS["Historic site grid",
    BASEPROJCRS["GDA94 / MGA zone 56",
        BASEGEOGCRS["GDA94",
            DATUM["Geocentric Datum of Australia 1994",
                ELLIPSOID["GRS 1980",6378137,298.257222101,
                    LENGTHUNIT["metre",1]]],
            PRIMEM["Greenwich",0,
                ANGLEUNIT["degree",0.0174532925199433]]],
        CONVERSION["Map Grid of Australia zone 56",
            METHOD["Transverse Mercator",
                ID["EPSG",9807]],
            PARAMETER["Latitude of natural origin",0,
                ANGLEUNIT["degree",0.0174532925199433],
                ID["EPSG",8801]],
            PARAMETER["Longitude of natural origin",153,
                ANGLEUNIT["degree",0.0174532925199433],
                ID["EPSG",8802]],
            PARAMETER["Scale factor at natural origin",0.9996,
```

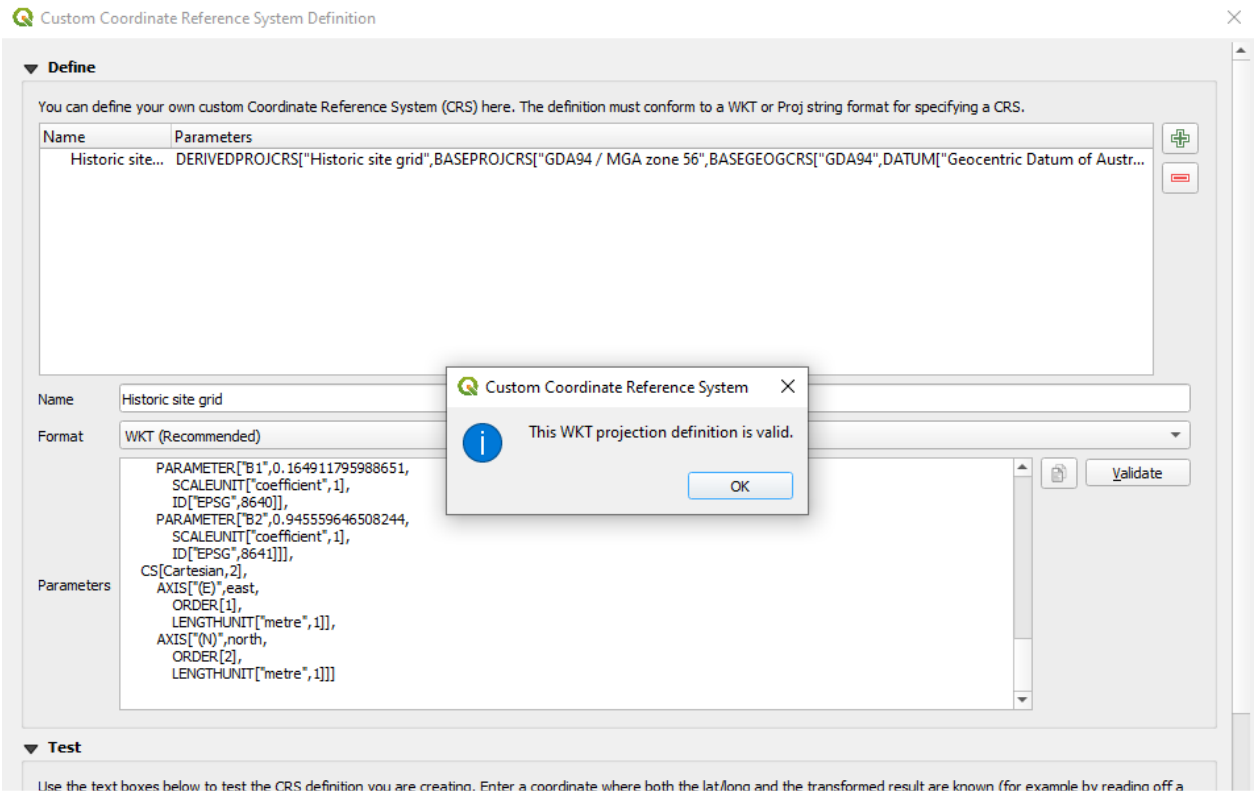
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```
                SCALEUNIT["coefficient",1],
```

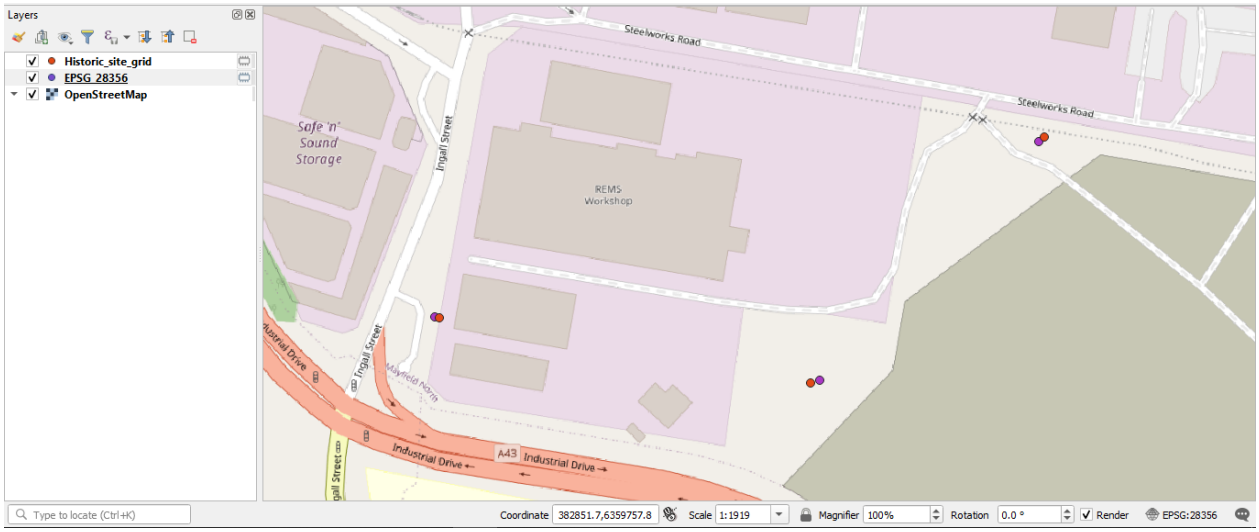
```
        ID["EPSG",8624]],
    PARAMETER["A2",-0.164911795988651,
        SCALEUNIT["coefficient",1],
        ID["EPSG",8625]],
    PARAMETER["B0",-6075863.374,
        LENGTHUNIT["metre",1],
        ID["EPSG",8639]],
    PARAMETER["B1",0.164911795988651,
        SCALEUNIT["coefficient",1],
        ID["EPSG",8640]],
    PARAMETER["B2",0.945559646508244,
        SCALEUNIT["coefficient",1],
        ID["EPSG",8641]]],
CS[Cartesian,2],
    AXIS["(E)",east,
        ORDER[1],
        LENGTHUNIT["metre",1]],
    AXIS["(N)",north,
        ORDER[2],
        LENGTHUNIT["metre",1]]]
```

That's all. Create the new Custom CRS:



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Share Improve this answer Follow edited Jul 2, 2020 at 16:01

answered Jul 2, 2020 at 15:55



Gabriel De Luca
13.1k 2 17 47

1 thank you @Gabriel De Luca. Great answer and it's given me a workable outcome out-of-the-box. If this was eBay I'd give you an A++++++++++ :) – timtim Jul 2, 2020 at 21:40

FWIW, I did have another shot at trying the multiple successive transformations, which I think is what @gabriel-de-luca you described in the other post (gis.stackexchange.com/questions/351394/...) making use of a PROJ pipeline. I feel I eventually understood each step in the process, but I couldn't make all the PROJ steps work together to give me a result. The WKT approach is accurate enough for 50 year old sketch maps! – timtim Jul 2, 2020 at 21:55

@timtim Hi, that is because we are defining the transformation from EPSG:28356 to the local cartesian one, to get a derived from projected CRS. You can set this CRS to the layer and export it to EPSG:28356, and the transformation applied is the inverse. If you want to transform the vector layer with ogr2ogr, you need to set a +inv parameter before the +proj=affine transformation, and the parameters are the same but named +xoff, +s11, +s12,+yoff, +s21 and +s22 instead of A1, A2, A3, B1, B2, B3, respectively. – Gabriel De Luca Jul 2, 2020 at 22:29

I also just want to thank you for this. I spent weeks trying to solve a similar problem. – Fiet Kleiner Apr 9, 2021 at 12:44

@FietKleiner, you are welcome – Gabriel De Luca Apr 9, 2021 at 13:19

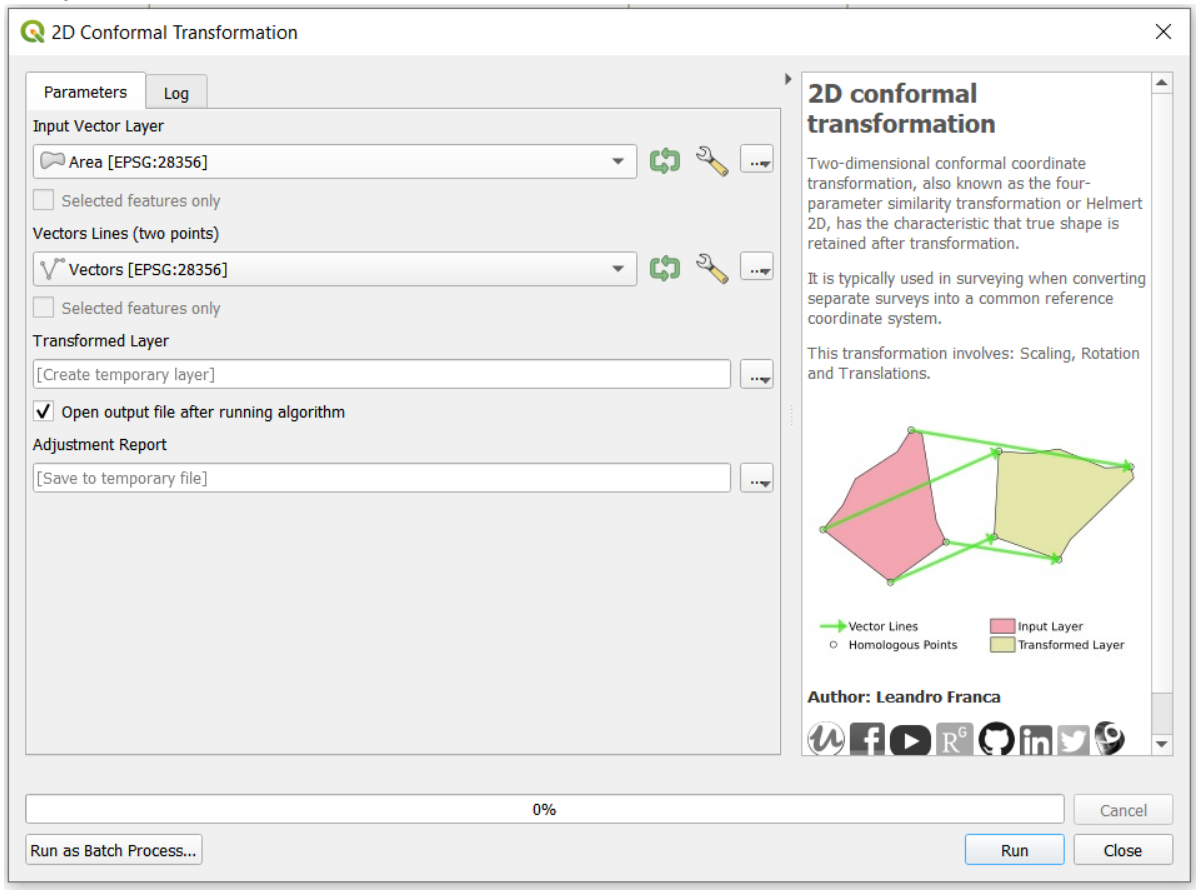
You should use the tool "2D conformal transformation" of the "LF Tools" QGIS plugin.

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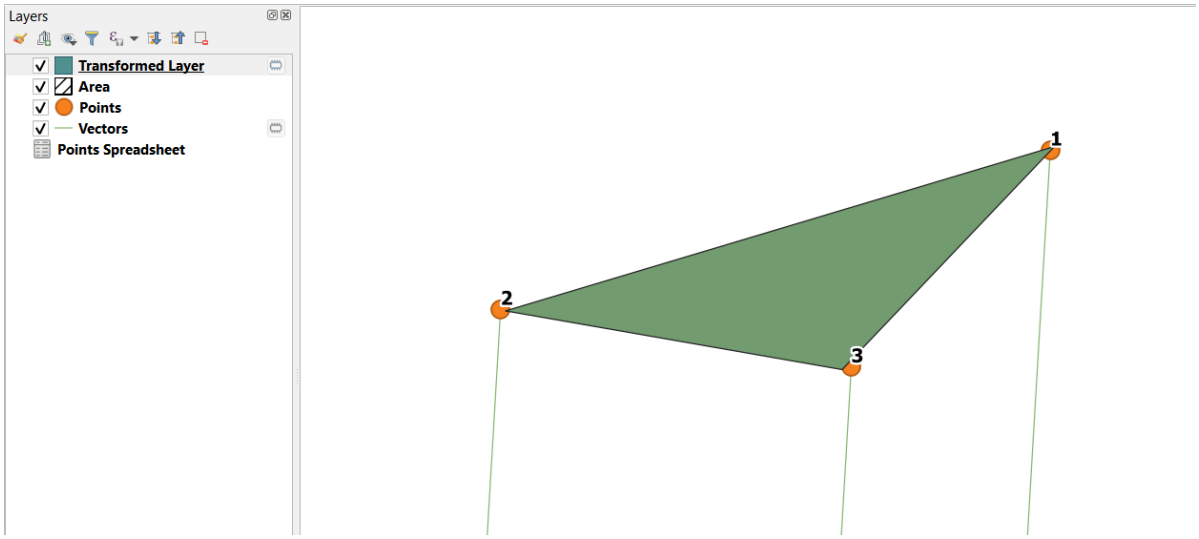
3. Lastly, run the tool ,as shown bellow:



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Look the result:



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answered May 23, 2021 at 18:25



Leandro França
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